SHREE H. N. SHUKLA COLLEGE OF I.T & MGMT.

(Affiliated to Saurashtra University & Gujarat Technological University)

Mobile Application Development in Android using Kotlin
BCA & BSC IT SEM 6

UNIT – 1 INTRODUCTION TO KOTLIN PROGRAMMING

Q- 1 Explain Basics of Kotlin.

❖ What is Kotlin?

- Kotlin is a new open source programming language.
- It like Java, JavaScript, Python etc.
- It is a high level strongly statically typed language.
- It combines functional and technical part in a same place.
- Currently, Kotlin mainly targets the Java Virtual Machine (JVM), but also compiles to JavaScript.
- Kotlin is motivated by other popular programming languages such as Java, C#, JavaScript.
- The syntax of Kotlin may not be exactly similar to Java Programming Language.
- internally Kotlin is dependent on the existing Java Class library to produce wonderful results for the programmers.
- Kotlin provides code safety, and clarity to the developers around the world.
- Kotlin 1.o was developed and released by JetBrains in 2016.
- Kotlin is free.
- It is developed under the Apache 2.0 license and the source code is available on GitHub

❖ Why Kotlin?

- Kotlin is getting high popularity among all level of programmers and it is used for:

  - ✓ Cross-platform Mobile applications.
  - ✓ Android Application Development.
  - ✓ Web Application Development
  - ✓ Server Side Applications

- ✓ Desktop Application Development
- ✓ Data science based applications

- Kotlin works on different platforms (Windows, Mac, Linux, etc.) and it's 100% compatible with Java.
- Kotlin is used by many large companies like Google, Netflix, Slack, Uber etc to develop their Android based applications.
- The most importantly, there are many companies actively looking for Kotlin developers, especially in the Android development space.

Q- 2 Explain Kotlin Operators.

- An operator is a symbol that tells the compiler to perform specific mathematical or logical manipulations.
- Kotlin is rich in built-in operators and provide the following types of operators:
  1. Arithmetic Operators
  2. Relational Operators
  3. Assignment Operators
  4. Unary Operators
  5. Logical Operators

1. Arithmetic Operators

- Kotlin arithmetic operators are used to perform basic mathematical operations such as addition, subtraction, multiplication and division etc.

Operator	Name	Description	Example
+	Addition	Adds together two values	x + y
-	Subtraction	Subtracts one value from another	x - y
*	Multiplication	Multiplies two values	x * y

/	Division	Divides one value by another	x / y
%	Modulus	Returns the division remainder	x % y

2. Relational Operators

- Kotlin relational (comparison) operators are used to compare two values, and returns a **Boolean** value: either **true** or **false**.

Operator	Name	Example
>	greater than	x > y
<	less than	x < y
>=	greater than or equal to	x >= y
<=	less than or equal to	x <= y
==	is equal to	x == y
!=	not equal to	x != y

3. Assignment Operators

- Kotlin assignment operators are used to assign values to variables.
- Following is an example where we used assignment operator = to assign a values into two variables:

Operator	Example	Expanded Form
=	x = 10	x = 10
+=	x += 10	x = x + 10
-=	x -= 10	x = x - 10

*=	x *= 10	x = x * 10
/=	x /= 10	x = x / 10
%=	x %= 10	x = x % 10

4. Unary Operators

- The unary operators require only one operand; they perform various operations such as incrementing/decrementing a value by one, negating an expression, or inverting the value of a boolean.

Operator	Name	Example
+	unary plus	+x
-	unary minus	-x
++	increment by 1	++x
--	decrement by 1	--x
!	inverts the value of a Boolean	!x

5. Logical Operators

- Kotlin logical operators are used to determine the logic between two variables or values:
- Following is the list of Kotlin Logical Operators:

Operator	Name	Description	Example
&&	Logical and	Returns true if both operands are true	x && y

||	Logical or	Returns true if either of the operands is true	x || y
!	Logical not	Reverse the result, returns false if the operand is true	!x

6. Bitwise Operations
   - Kotlin does not have any bitwise operators but Kotlin provides a list of helper functions to perform bitwise operations.

Q- 3 Explain Kotlin Variables.

- Variables are containers for storing data values.
- To create a variable, use **var or val**, and assign a value to it with the equal sign (=):
- **Syntax**
- var *variableName* = *value*
- val *variableName* = *value*
- **Example:-**

```
var name = "John"
val birthyear = 1975

println(name)        // Print the value of name
println(birthyear)   // Print the value of birthyear
```

Q- 5 Explain Kotlin Data types.

- Kotlin has many types of data types likes
- Numbers
- Characters
- Booleans
- Strings
- Arrays

Numbers

- Number types are divided into two groups:
- **Integer types** store whole numbers, positive or negative (such as 123 or -456), without decimals. Valid types are Byte, Short, Int and Long.
- **Floating point types** represent numbers with a fractional part, containing one or more decimals. There are two types: Float and Double.

Characters

- The Char data type is used to store a **single** character. A char value must be surrounded by **single** quotes, like 'A' or 'c':

String:-

- The String data type is used to store a sequence of characters (text). String values must be surrounded by **double** quotes:

Array

- Arrays are used to store multiple values in a single variable, instead of declaring separate variables for each value.

Booleans:-

- Very often, in programming, you will need a data type that can only have one of two values, like:

- YES / NO
- ON / OFF
- TRUE / FALSE

Q- 6 Explain Kotlin Decision Making.

- There are case where we want a block of code to be executed when some condition is satisfied.
- In Kotlin, we have Decision Making Statement that enable computer to decide which block of code to be execute based on some conditional choices. Decision making statement statements is also referred to as selection statements.

STATEMENT	DESCRIPTION
if Statements	Block of statement executed only when specified test expression is true.
if else Statements	When we want to execute some block of code if a condition is true and another block of code if a condition is false, In such a case we use if….else statement.
if..else..if Ladder	It tests set of conditions in sequence.
Nested Statements	When there is an if statement inside another if statement then it is known as nested if else.

✓ **KOTLIN - IF...ELSE EXPRESSION**

- Kotlin **if...else** expressions works like an **if...else** expression in any other Modern Computer Programming. So let's start with our traditional if...else statement available in Kotlin.
- **Syntax**
- The syntax of a traditional **if...else** expression is as follows:

```
if (condition) {
  // code block A to be executed if condition is true
} else {
  // code block B to be executed if condition is false
}
```

- Here **if** statement is executed and the given **condition** is checked. If this condition is evaluated to **true** then code block A is executed, otherwise program goes into else part and code block B is executed.

- **Syntax**

```
val result = if (condition) {
  // code block A to be executed if condition is true
} else {
  // code block B to be executed if condition is false
}
```

- If you're using if as an expression, for example, for returning its value or assigning it to a variable, the else branch is mandatory.

➢ **Kotlin if...else...if**

- You can use **else if** condition to specify a new condition if the first condition is false.
- **Syntax**

```
if (condition1) {
```

```
  // code block A to be executed if condition1 is true
} else if (condition2) {
  // code block B to be executed if condition2 is true
} else {
  // code block C to be executed if condition1 and condition2 are false
}
```

- **Kotlin Nested if Expression**
  - Kotlin allows to put an if expression inside another if expression. This is called nested if expression
  - **Syntax**

```
if(condition1) {
   // code block A to be executed if condition1 is true
   if( (condition2) {
      // code block B to be executed if condition2 is true
   }else{
      // code block C to be executed if condition2 is fals
   }
} else {
  // code block D to be executed if condition1 is false
}
```

Q- 5 Explain Kotlin Loop Control.

1. Kotlin while Loop

- The **while loop** is used to iterate a part of program several time. Loop executed the block of code until the condition has true.
- Kotlin while loop is similar to Java while loop.
- **Syntax**

```
while(condition)
{
    //body of loop
}
```

2. Kotlin do-while Loop

- The do-while loop is similar to while loop except one key difference. A do-while loop first execute the body of do block after that it check the condition of while.
- As a do block of do-while loop executed first before checking the condition, do-while loop execute at least once even the condition within while is false.
- The while statement of do-while loop end with ";" (semicolon).

- Syntax

```
do
{
    //body of do block
}
while(condition);
```

3. Kotlin for Loop

- Kotlin for loop is used to iterate a part of program several times.
- It iterates through arrays, ranges, collections, or anything that provides for iterate.
- Kotlin for loop is equivalent to the foreach loop in languages like C#.
- Syntax:

```
for (item in collection)
{
  //body of loop
}
```

Q- 5 Explain Kotlin Collections.

- Collections are a common concept for most programming languages.
- A collection usually contains a number of objects of the same type and Objects in a collection are called elements or items.
- The Kotlin Standard Library provides a comprehensive set of tools for managing collections. The following collection types are relevant for Kotlin:

1. **Kotlin List -** List is an ordered collection with access to elements by indices. Elements can occur more than once in a list.
2. **Kotlin Set -** Set is a collection of unique elements which means a group of objects without repetitions.
3. **Kotlin Map -** Map (or dictionary) is a set of key-value pairs. Keys are unique, and each of them maps to exactly one value.

Kotlin Collection Types

- Kotlin provides the following types of collection:
  1. Collection or Immutable Collection
  2. Mutable Collection

    1. Kotlin Immutable Collection

- Immutable Collection or simply calling a Collection interface provides read-only methods which means once a collection is created, we can not change it because there is no method available to change the object created.

Collection Types	Methods of Immutable Collection
List	listOf() listOf<T>()
Map	mapOf()
Set	setOf()

Example

```
fun main()
{
    val numbers = listOf("one", "two", "three", "four")
    println(numbers)
}
```

2. Kotlin Mutable Collection

- Mutable collections provides both read and write methods.

Collection Types	Methods of Immutable Collection
List	ArrayList<T>() arrayListOf() mutableListOf()
Map	HashMap hashMapOf() mutableMapOf()
Set	hashSetOf() mutableSetOf()

Q- 6 Explain Kotlin Functions.

- A function is a block of code which only runs when it is called.
- You can pass data, known as parameters, into a function.
- Functions are used to perform certain actions, and they are also known as methods.
- There are two types of functions depending on whether it is available in standard library or defined by user.
    1. Standard library function
    2. User defined function

1. **Standard library function**

- Kotlin Standard library function is built-in library functions which are implicitly present in library and available for use.

2. **User defined function**

- User defined function is a function which is created by user. User defined function takes the parameter(s), perform an action and return the result of that action as a value.
- Kotlin functions are declared using the fun keyword. For example:

```
fun functionName()
{
    // body of function
}
```

- We have to call the function to run codes inside the body of the function.

```
functionName()
```

➢ **Kotlin Recursion Function**

- Recursion function is a function which calls itself continuously. This technique is called recursion.
- **Syntax**

```
fun functionName()
{
.. .. ..
functionName() //calling same function
}
```

- **Kotlin Default Argument**

- Kotlin provides a facility to assign *default argument* (parameter) in a function definition.
- If a function is called without passing any argument than default argument are used as parameter of function definition.

- **Kotlin Lambda Function**

- Lambda is a function which has no name.
- Lambda is defined with a curly braces **{}.**
- which takes *variable as a parameter* (if any) and body of function.

Syntax of lambda

{ variable -> body_of_function}

- **Inline Function**

- An **inline function** is declare with a keyword inline.
- The use of inline function enhances the performance of higher order function.
- The inline function tells the compiler to copy parameters and functions to the call .
- The virtual function or local function cannot be declared as **inline**.

Q- 7 Kotlin – What is OOP?

- OOP stands for Object-Oriented Programming.
- Procedural programming is about writing procedures or methods that perform operations on the data, while object-oriented programming is about creating objects that contain both data and methods.
- Object-oriented programming has Many advantages over procedural programming:
    - OOP is faster and easier to execute.
    - OOP provides a clear structure for the programs.

- OOP helps to keep the Kotlin code makes the code easier to maintain, modify and debug.
- OOP makes it possible to create full reusable applications with less code and shorter development time.

➢ **Kotlin Classes**

- A class is a blueprint for the objects which defines a template to be used to create the required objects.
- Classes are the main building blocks of any Object Oriented Programming language.
- A Kotlin class is defined using the class keyword. Following is the syntax to create a Kotlin Class:

```
class ClassName
{
    // Class Header
     // Variables or data members
   // Member functions or Methods
  ...
}
```

- By default, Kotlin classes are public.

Q- 8 Explain Kotlin visibility Modifier.

- **Visibility modifiers** are the keywords which are used to restrict the use of class, interface, methods, and property of Kotlin in the application.
- These modifiers are used at multiple places such as class header or method body.
- In Kotlin, visibility modifiers are categorized into four different types:
    1. public
    2. protected

3. internal
4. private

1. **public modifier**

- A **public** modifier is accessible from everywhere in the project. It is a default modifier in Kotlin.
- If any class, interface etc. are not specified with any access modifier then that class, interface etc. are used in public scope.

2. **protected modifier**

- A **protected** modifier with class or interface allows visibility to its class or subclass only.

3. **internal modifier**

- The internal modifiers are newly added in Kotlin.
- it is not available in Java.
- Declaring anything makes that field marked as internal field.
- The internal modifier makes the field visible only inside the module in which it is implemented.

4. **private modifier**

- A **private** modifier allows the declaration to be accessible only within the block in which properties, fields, etc. are declare.
- The private modifier declaration does not allow to access the outside the scope.
- A private package can be accessible within that specific file.

Q- 9 Kotlin – What is Inheritance?

- Inheritance is an important feature of object oriented programming language.

- Inheritance allows to inherit the feature of existing class (or base or parent class) to new class (or derived class or child class).
- The main class is called super class (or parent class) and the class which inherits the superclass is called subclass (or child class).
- The subclass contains features of superclass as well as its own.
- The concept of inheritance is allowed when two or more classes have same properties
- It allows code reusability.
- A derived class has only one base class but may have multiple interfaces whereas a base class may have one or more derived classes.
- In Kotlin, the derived class inherits a base class using: operator in the class header (after the derive class name or constructor)

```
open class Base(p: Int)
{
  -----
}
class Derived(p: Int) : Base(p)
{
  -----
}
```

➢ **Kotlin open keyword**

- As Kotlin classes are **final** by default, they cannot be inherited simply. We use the **open** keyword before the class to inherit a class and make it to non-final.

Q- 10 Kotlin – What is abstract class?

- A class which is declared with abstract keyword is known as abstract class.
- An abstract class cannot be instantiated.
- Means, we cannot create object of abstract class.
- The method and properties of abstract class are non-abstract unless they are explicitly declared as abstract.

Declaration of abstract class

```
abstract class A
{
    var x = 0
   abstract fun doSomething()
}
```

- Abstract classes are partially defined classes, methods and properties which are no implementation but must be implemented into derived class.
- If the derived class does not implement the properties of base class then is also meant to be an abstract class.
- Abstract class or abstract function does not need to annotate with open keyword as they are open by default.
- Abstract member function does not contain its body.
- The member function cannot be declared as abstract if it contains in body in abstract class.

Q- 11 Explain super and this keyword in details.

➢ **Super Keyword**

- The super keyword in kotlin is a reference variable which is used to refer immediate parent class object.

- Whenever you create the instance of subclass, an instance of parent class is created implicitly which is referred by super reference variable.
- Usage of kotlin super Keyword
  1. super can be used to refer immediate parent class instance variable.
  2. super can be used to invoke immediate parent class method.
  3. super() can be used to invoke immediate parent class constructor.

➢ **this Keyword**

- The this keyword refers to the current object in a method or constructor.
- The most common use of the this keyword is to eliminate the confusion between class attributes and parameters with the same name (because a class attribute is shadowed by a method or constructor parameter).
- this can also be used to:
  1. Invoke current class constructor
  2. Invoke current class method
  3. Return the current class object
  4. Pass an argument in the method call
  5. Pass an argument in the constructor call

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Mobile Application Development in Android using Kotlin
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UNIT – 2 Introduction to Android & Android Application Design

- The Open Handset Alliance
- The Android Platform, Android SDK
- Building a sample Android application
- Anatomy of an Android applications
- Android terminologies
- Application Context, Activities, Services, Intents
- Receiving and Broadcasting Intents
- Android Manifest File and its common settings
- Using Intent Filter, Permissions
- Managing Application resources in a hierarchy Working with different types of resources

Handset Manufacturers

- Before Android, mobile developers faced many problem when it came to writing application.
- Building the better application, the unique application, the competing application, the hybrid application and incorporating many common tasks such as messaging and calling in a familiar way were often unworkable goals.

First Generation Mobile

- These early phones were some problem, but they did something important – They changed the way about communication.
- As mobile phone prices dropped, batteries improved, and reception areas grew, more and more people began carrying these handy devices.

WAP(Wireless Application Protocol)

- Wireless Application Protocol (WAP) is a technical standard for accessing information over a mobile wireless network
- A WAP browser is a web browser for mobile devices such as mobile phones that use the protocol.
- User filled out a simple order form with his phone number and his handset model.
- It was up to the content provider to deliver an image or audio file compatible with the given phone.
- Payment and verification were handled through various premium priced delivery mechanisms such as Short Message Service (SMS), Enhanced Messaging Service (EMS), and Multimedia Messaging Service (MMS).

Different Platforms

- A variety of different platforms emerged—and developers are still actively creating applications for them.
    - Some smartphone devices ran Palm OS (now Garnet OS) and RIM BlackBerry OS.

- Sun Microsystems took its popular Java platform and J2ME emerged (now known as Java Micro Edition [Java ME]).
- Chipset maker Qualcomm developed and licensed its Binary Runtime Environment for Wireless (BREW).
- Other plat- forms, such as Symbian OS, were developed by handset manufacturers such as Nokia,Sony Ericsson, Motorola, and Samsung.
- The Apple iPhone OS (OS X iPhone) joined the ranks in 2008.

The Open Handset Alliance.(જોડાણ)

- The **Open Handset Alliance (OHA)** is a consortium of 84 firms to develop open standards for mobile devices.
- Members include HTC, Sony, Dell, Intel, Motorola, Qualcomm, Texas Instruments, Google, Samsung Electronics, LG Electronics, T-Mobile, Sprint Corporation (now merged with T-Mobile US), Nvidia, and Wind River Systems.
- The Open Handset Alliance (OHA) was formed in November 2007.
- The OHA is a business alliance comprised(સમાવેશ) of many of the largest and most successful mobile companies on the planet.
- Its members include chip makers, handset manufacturers, software developers, and service providers.
- The entire mobile supply chain is well represented.
- Google was found this alliance.
- Head quarter at **Mountain View, California, United States .**
- The Open Handset Alliance is a group of mobile and technology leaders who share this vision for changing the mobile experience for consumers.
- **Andy Rubin** has been credited as the father of the Android platform.
- His company, Android Inc., was **acquired (Purchased) by Google in 2005**.
- Working together, OHA members, including Google, began developing a non-proprietary(માલિકી) open standard platform based

upon technology developed at Android Inc.

- that would aim to alleviate(improve) the main problems comes the mobile community.
- The result is the Android project. To this day, most android platform development is completed by Rubin's team at Google, where he acts as VP of Engineering and manages the Android platform roadmap.
- Google hosts the Android open source project and provides online Android documentation, tools, forums, and the Software Development Kit (SDK) for developers. All major Android news originates at Google.

Sr.no	Questions	Answer
1	OHA full form is____	Open Handset Alliance
2	OHA formed in	November 2007
3	WAP full form is_	Wireless Application Protocol

The Android Platform

- The Android Platform was launched in 2007 by the Open Handset Alliance(Jodan), an alliance of famous companies that includes Google, HTC, Motorola and others.
- Although most of the applications that run on the Android Platform are written in Java, there is no Java Virtual Machine.
- Instead, the Java classes are first compiled into what are known as Dalvik Executable and run on the Dalvik Virtual Machine. Pre-commercial release versions Alpha -> Release November 2007 Beta -> Release Nov 5, 2007. Latest version of android with logo.

Figure 1.5 The Android mascot and logo.

Figure 1.6 Some Android SDKs and their codenames.

Features of android:-

- **Free and Open Source**
- **Familiar and Inexpensive Development Tools**
- **Freely Available Software Development Kit**
- **Familiar Language, Familiar Development Environments**
  - Windows XP (32-bit) or Vista (32-bit or 64-bit)
  - Mac OS X 10.5.8 or later (x86 only)
  - Linux (tested on Linux Ubuntu 8.04 LTS, Hardy Heron)
- **A "Free Market" for Applications**

- Android developers can distribute their applications to users in a variety of ways:

Android SDK

- The Android Software development kit (SDK).
- includes a comprehensive set of development tools.
- These include a debugger, libraries, a handset emulator based on Quick Emulator, documentation, sample code, and tutorials
- It use java programming
- The officially supported integrated development environment (IDE) is Eclipse using the Android Development Tools (ADT) Plugin.
- But now Android Studio is the official IDE;
- Note : You must install java 1.6 or latter in order to work on android

- What is Android?

The Android operating system

- ✓ Android is a mobile operating system based on a modified version of the Linux kernel and other open-source software, designed primarily for touchscreen mobile devices such as smartphones and tablets.
- ✓ Android is an operating system based on the Linux kernel.
- ✓ The first commercial Android device the HTC Dream, being launched in September 2008.
- ✓ Over 70 percent of smartphones based on Android
- ✓ Android written in c. c++, java , Kotlin and many other.
- ✓ It available in 100 + languages.
- ✓ Latest version of android is android 14, released on 4th October-23
- ✓ The project responsible for developing the Android system is called the **Android Open Source Project** (AOSP) and is primarily lead by Google.

- ✓ The Android system supports background processing, provides a rich user interface library, supports 2-D and 3-D graphics using the OpenGL-ES (Open Graphics Library for Embedded Systems (short OpenGL) standard and grants access to the file system as well as an embedded SQLite database.

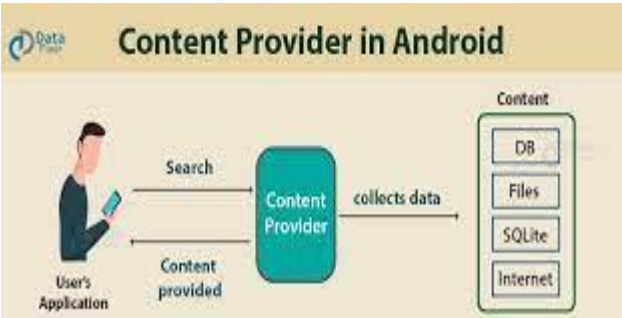
- ✓ An Android application typically consists of different visual and non visual components and can reuse components of other applications.

Anatomy(રચના) of an Android applications

- Android application is developed by one or more components that work together, these components are activities.
- Activity is a single and independent module that is correlated to the application's single screen and its functionalities
- There are four main building block to android application likes

- Activity.
- Intent receiver(intent)
- Services
- Content provider.

1) **Intent** :-
   - ➔ Intent is messaging object used to request any action from another application component.
   - ➔ Intents facilitate communication between different components in many ways.
   - ➔ Example of intent are gallery, browser, contact many other tab in our mobile.
2) **Services**:-
   - ➔ A Service is an application component that can perform long-running operations in the **background.** It does not provide a user interface.
3) **Content provider:-**
   - ➔ Content providers are the standard interface that connects data in one process with code running in another process.
   - ➔ Implementing a content provider has many advantages.

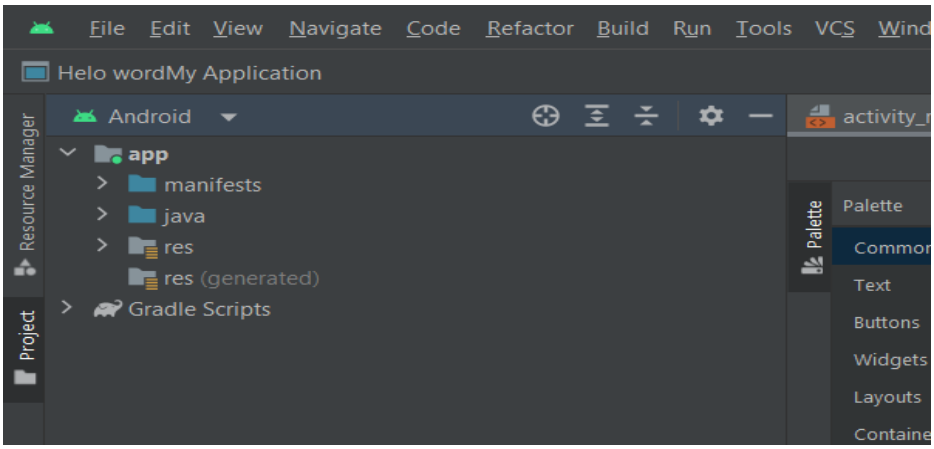


Some of other anatomy (framer or body of android are as follow.

- Src : Java or kotlin source code files will be available here.\

- Res : res used to store the values for resources that are used in many android project to include color , style andy many other.

- it is folder available in android studio.
  - /res/values : Used to define strings, colors, dimensions, styles, static arrays etc. Ex. String are defined in the res/values/string.xml file.
  - /res/layout : This folder contains the layouts to be used in the application.
  - /res/menu : This folder contains menu resources to be used in the application
  - /res/drawable : 1) Drawable resource is general concept for a graphics that can be drawn to the screen.
    2) Drawable folders are resource directories in the application that provides differrent bitmap drawables for medium, high and extra high solidity screens.



Explain Android terminologies

1) apk file :

➔ Android application package file.

➔ **Android package kit file.**

➔ it is file format for application used on android os

2) dex file :

➔ Delvik Executable File that all application must have.

➔ It Compiled Android application code file.

➔ .dex files can be created by automatically translating compiled applications written in the **java programming language.**

3) **Action** :
An action is a string value assigned to Intent.

4) **Activity** :
➔ It is specifies combination of xml file and java files, Kotlin file.
➔ It combine coding and design

5) **Adb :**
➔ Android Debug Bridge, a command line debugging application includes with the SDK.
➔ It is versatile command line tool that communicate with device. Such as installing and debugging app.

6) **Canvas** : Canvas is the simplest, easiest way to draw 2D objects on the screen.

7) **Content Provider :** A content provider is built in the ContentProvider class. It is used to store and retrieve data such as contact, images, videos ,music other media.

8) **Dalvik:**
➔ It is discontinue process virtual machine in android os..
➔ The Android Platform's virtual machine.
➔ The Dalvik VM is an interpreter only VM that executes files in the Dalvik Executable (.dex) format.
➔ It not used in newer version of android.

9) **DDMS** : Dalvik Debug Monitor Service, a GUI debugging application included with the SDK.

10) **Dialog** : dialog is small window that prompts the user to make decision or enter additional Information. It is simple button like checkbox radio button.

11) **Drawable** : this resources is general concept for graphics that can be drawn to the screen.

12) **Layout Resources** : An XML file that describes that layout of an Activity screen

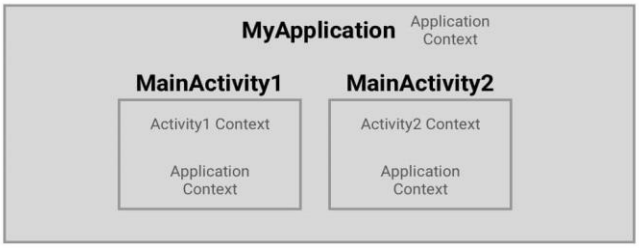
13) **Manifest File :** An XML file that each application must define, to describe the app's package name, version, component etc.

14) **XML**:in android, xml is used for designing the application's user interface(UI)like creating layout ,view ,buttons ,text fields etc.

15) **Emulator**:- it is android virtual device , here we can run and test our project .

What is Application Context?

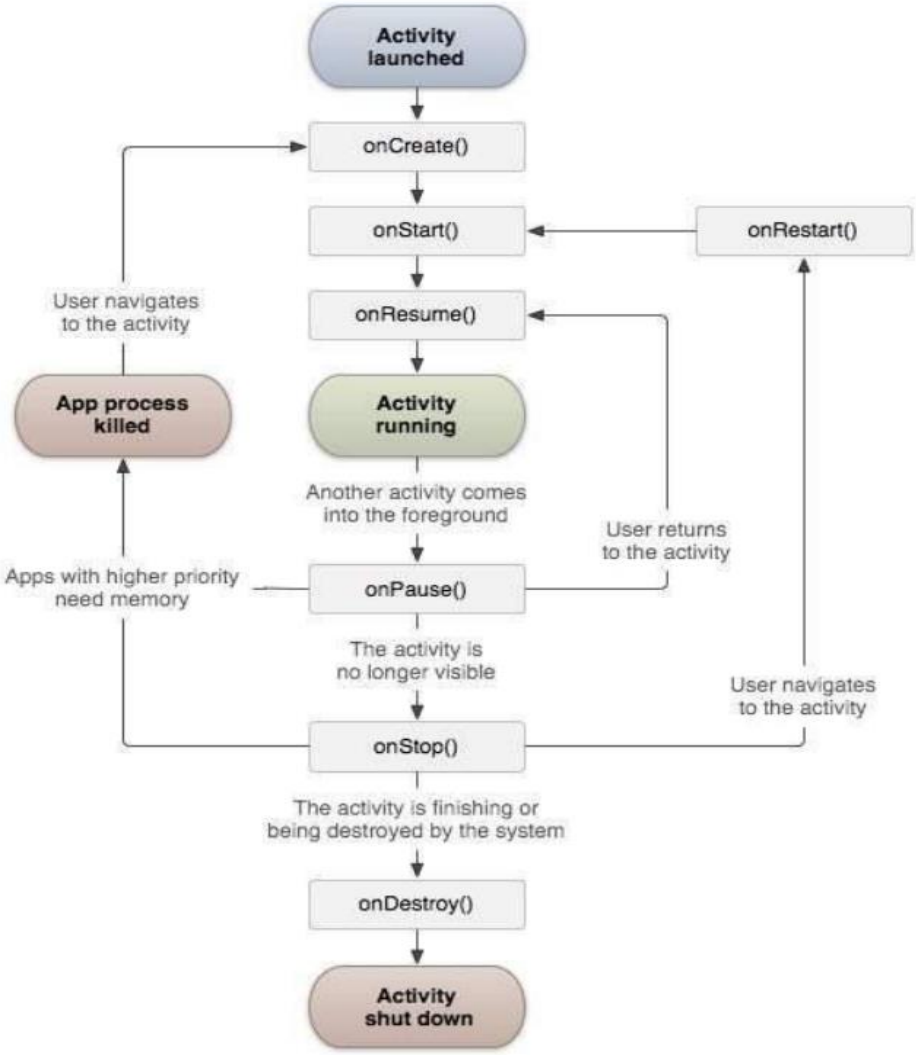
- Context is the interface that contain fully detail information about application or your project.
- Context is allow to access string , themes, color and many other.
- It used to application-level operations such as launching activities, broad casting and receiving intents.
- It is first step or first page of our project.



What is Activity?:-

- If you have worked with C, C++ or Java programming language then you must have seen that your program starts from main() function.
- Very similar way, Android system start its program with Activity.
- It is specifies combination of xml file and java files, Kotlin file.
- It combine coding and design
- The xml file provide the design of the screen and java or Kotlin file deal with coding
- Activity consist of two things like front end and back end .

- **Activity life cycle diagram(Explain activity lifecycle in detail):**



- **<u>Main components of lifecycle:-</u>**

  (i) **<u>onCreate():-</u>** This is the first steps. It is called when the activity is first created.

  (ii) **<u>onStart()</u>**:- This method is called when the activity becomes visible to the user.

  (iii) **onResume():**This method is called when the user starts interacting with the application.

  (iv) **<u>onPause()</u>** it called when activity is not visible to the user.

  (v) **<u>onStop()</u>** This method is called when the activity is no longer visible.

(vi) **onDestroy()** This method is called before the activity is destroyed by the system.

(vii) **onRestart()** This method is called when the activity restarts after stopping it.

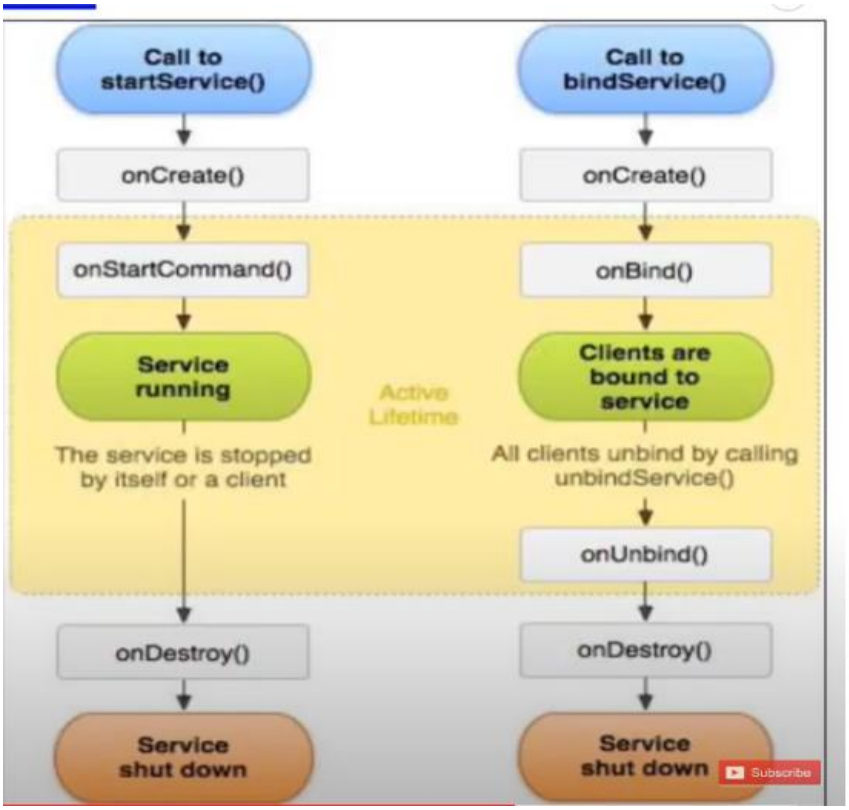
- **Explain Service in detail.**
  - ➔ A service is a component that runs in the background to perform long-running operations without needing to interact with the user and it works even if application is destroyed.
  - ➔ For example, a service might play music in background while user in a different application, or it might fetch data over network without blocking user interface with an activity.
  - ➔ A service can necessary take two states likes **started and bound**

1) Started:-

- A service is started when an application component, such as an activity, starts it by calling startService().used for only for single activity.
- Once started, a service can run in the background .

2) Bound:-

- A service is bound when an application component binds to it by calling bindService().used for two or more activity
- A bound service offers a client-server interface that allows components to interact with the server, send requests, get results.



- The diagram on left show the lifecycle when the services is created with **startservice()**.it used only for one activity .
- The diagram on right show the lifecycle when the services is created with **bindservice()**.it is used more then one activity.

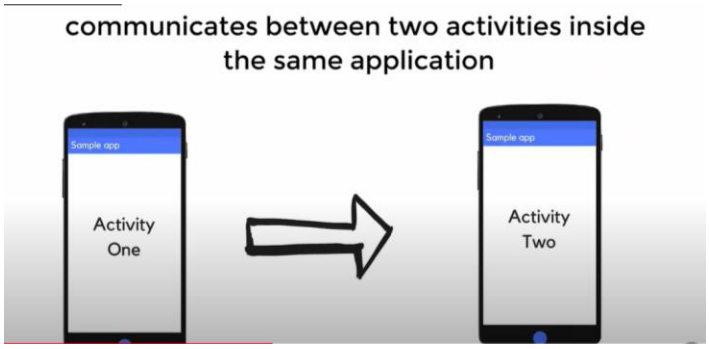
➔ Explain Intents:-

- an intent is to perform an action on the screen.
- Intent are used to start activity.
- Start a services(open email , web browser ,& calling).
- It also used for broadcast messaging.
- Pass data in same application or different application.
- There are two intents available in android as Implicit Intents and Explicit Intents.

Types of Intents:-

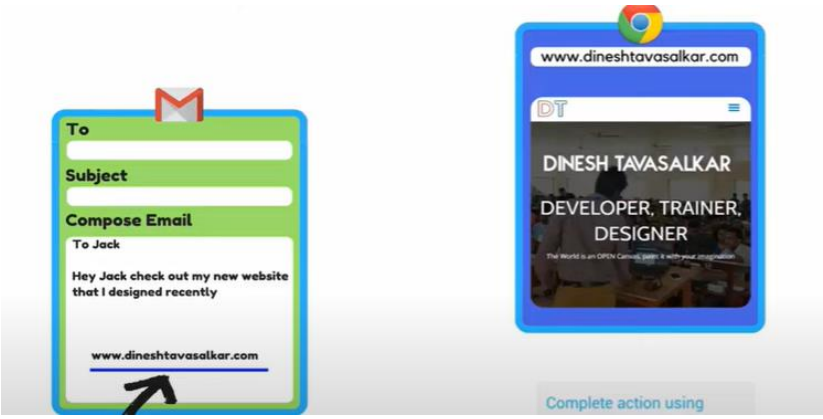
1) Explicit Intent:-

- Communication between two activities inside the same application.



2) Implicit Intents:-

- Communication between two activities of different application.



Explain Receiving and Broadcasting Intents:-

- Broadcast in android is the system events.
- It can occur when the device is start or msg is receive on device.
- It also occur when incoming call are received or device are goes to airplane mode.
- It is used to response the system event.
- Thera are two type of broadcast receiver like 1) Static Broadcast 2) Dynamic broadcast receiver.

1) Static broadcast receiver:-

- These type of receiver are declare in the manifest file and works even If the app is closed.

2)Dynamic broadcast receiver:-

- These type of receiver are works only if the app is running or minimize.

- **Some the important intent generate by system are as follow**

1) android.intent.action.BATTERY_CHANGED:-

- it indicate that changed the battery.

2) android.intent.action.BATTERY_Low:-

- Indicates low battery condition on the device.

3) android.intent.action.BATTERY_OKAY:-

- Indicates the battery is now okay after being low.

4) android.intent.action.BOOT_COMPLETED:-

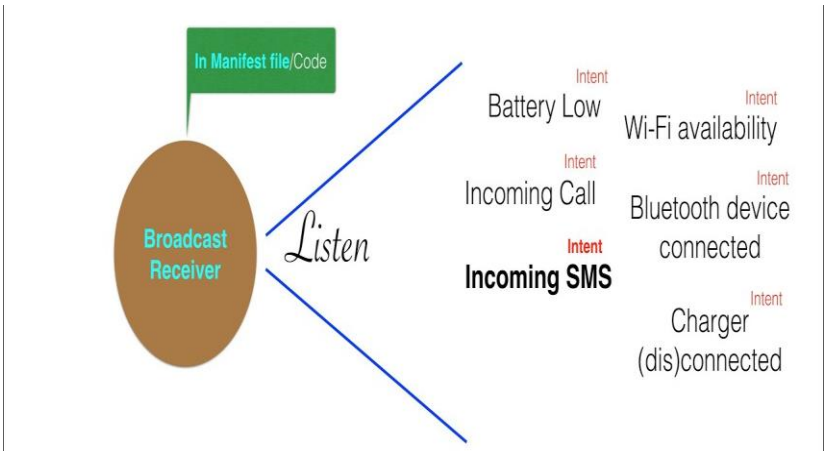
- This is broadcast once, after the system has finished booting.

5) android.intent.action.BUG_REPORT :-

- Show activity for reporting a bug.

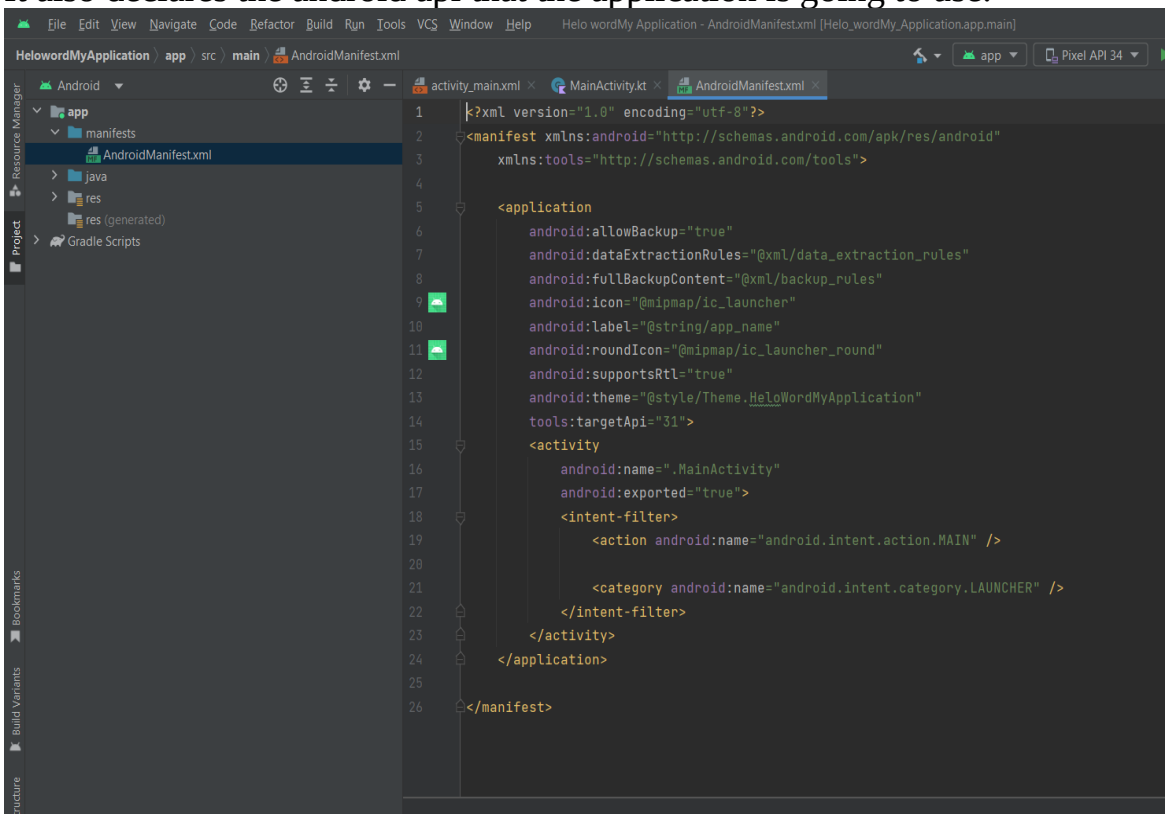
6) android.intent.action.REBOOT:-

- Have the device reboot.



Explain Android Manifest File and its common settings:-

- The Android Manifest is an XML file.
- It is simple text file that provide all the information about app.
- When we can create any app , default manifest file is created automatically.
- it contains important metadata about the Android app.
- This includes the package name, activity names, main activity, Android version support, hardware features support, permissions, and other configurations.
- The Android manifest file helps to declare the permissions.
- It also declares the android api that the application is going to use.



The elements used in the xml file are described below.(common setting in manifest file)

➔ <manifest>:-

- manifest is the root element of the AndroidManifest.xml file.

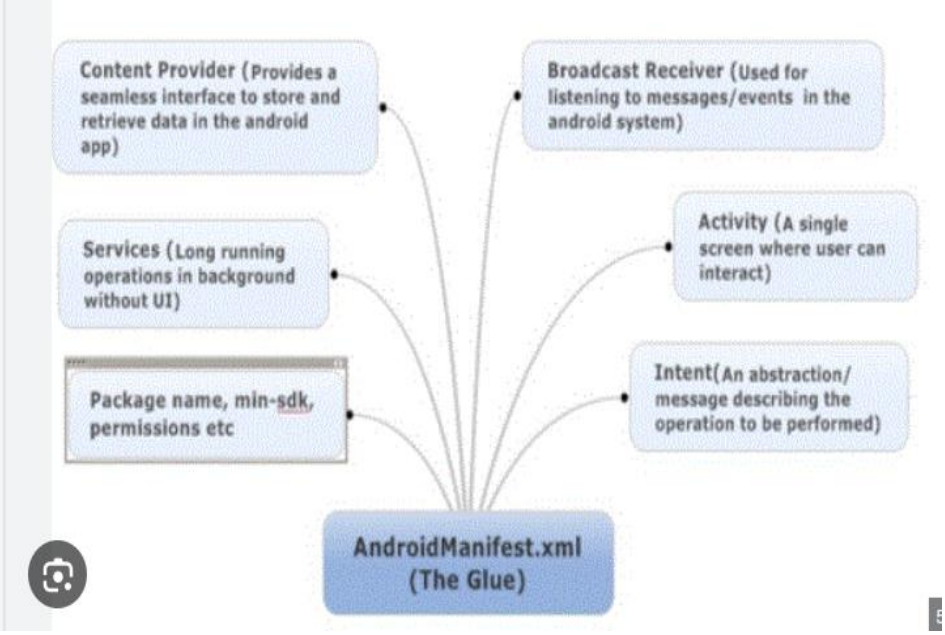
- It is define components of app.

➔ **<application>:-**

- application is the sub element of the manifest.
- It includes the namespace declaration.
- This element contains many sub elements.
- It contain app basic icon , icon label and themes for any app.
- The commonly used attributes are of this element are icon, label, theme etc.
  - **android:icon** represents the icon for all the android application components.
  - **android:label** works as the default label for all the application components.
  - **android:theme** represents a common theme for all the android activities.

➔ **Activity:-** activity is the sub element of application.

- It communication of main codding file like java or Kotlin.
- If activity can not declare then app is never run.

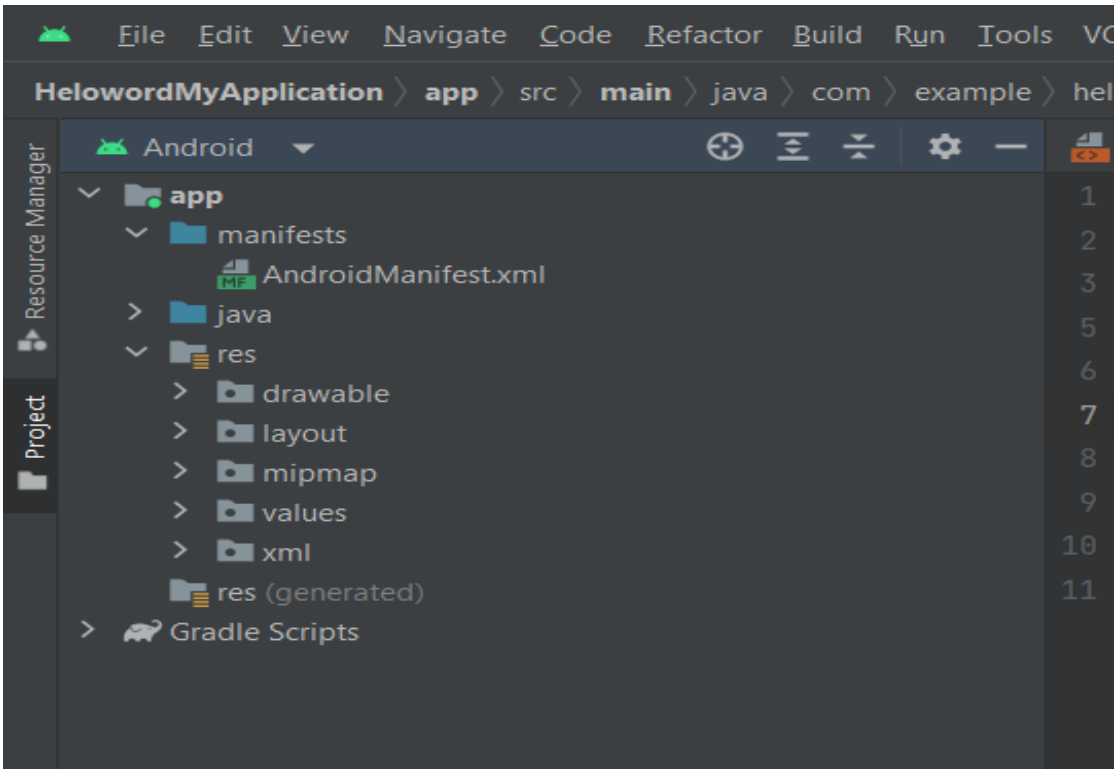


What is Resources in android ? explain in detail.

Or

Explain Managing Application resources in a hierarchy and Working with different types of resources:-

- There are many more items which we can use to build a good Android application.
- Resources are the additional files and static content that our code uses.
- such as bitmaps, colors, layout definitions, user interface strings, animation instructions, and more.
- These resources are always maintained separately in various sub-directories under res/ directory of the project.



➔ **anim/:-** XML files that define property animations. They are saved in res/anim/ folder.

➔ **color/:-** XML files that define a state list of colors. They are saved in res/color/ .

- **Drawable/ :-** Image files like .png, .jpg, .gif or XML files that are compiled saved in res/drawable/
- **Layout/ :-** XML files that define a user interface layout. They are saved in res/layout/

SHREE H. N. SHUKLA COLLEGE OF I.T & MGMT.

(Affiliated to Saurashtra University & Gujarat Technological University)

Mobile Application Development in Android using Kotlin
BCA & BSC IT SEM 6

UNIT – 3 Android User Interface Design

- User Interface Screen elements
  - ➔ Button, EditText, TextView, DatePicker, TimePicker, ProgressBar, ListView, GridView, RadioGroup, ImageButton, Fragement
- Designing User Interfaces with Layouts
  - ➔ Relative Layout, Linear Layout, Table Layout etc
- Dialogs
- Drawing and Working with Animation
- Twined Animation

Q. Explain Front end and back end in android.

Front End:-

- It is all about what a user can see.
- It's may be button, images, layout and input field and many more.
- To design these component in android we need XML and Android programming (Kotlin or Java)
- It is also called **UI**.

Back End:-

- Back end of anything is not visible to general user or administrator.
- It can be piece of code or program running on the server machine to serve the user need.
- Date base is also a part of backend (like SQLite).
- It is also called **Uact**

Q. Explain User interface overview.(UI):-

- ➔ The role of the UI is to display the application data on the screen and also to serve as the primary point of user interaction.
- ➔ Whenever the data changes, either due to user interaction (like pressing a button) or external input (like a network response), the UI should update to reflect those changes.

- **UI screen elements:**
  - ➔ TextView
  - ➔ EditText
  - ➔ AutoCompleteTextView
  - ➔ Button ImageButton
  - ➔ ToggleButton
  - ➔ RadioButton
  - ➔ RadioGroup
  - ➔ ProgressDialog
  - ➔ Spinner
  - ➔ Time Picker

- ➔ Date Picker
- ➔ ListView
- ➔ GridView
- ➔ Fragment.

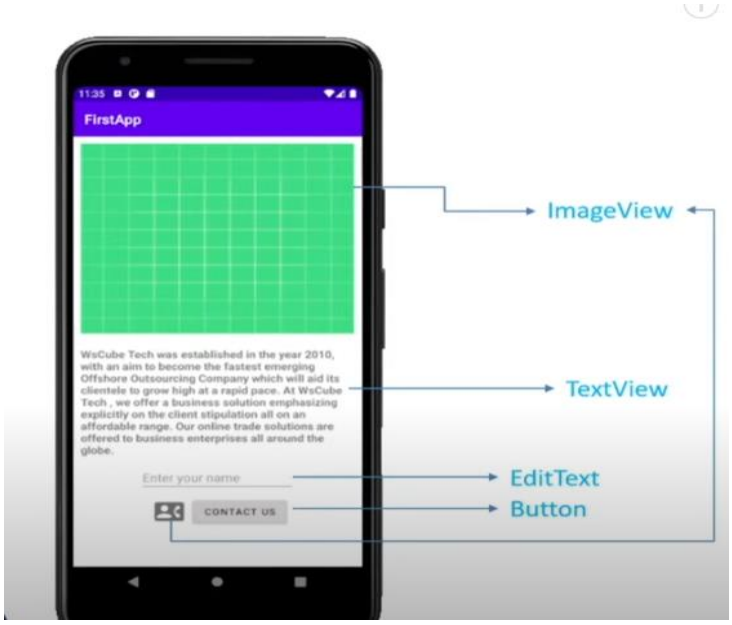
Note:- android Studio Download and install full step by step video link please watch this video carefully and then install android studio.

Link:- https://www.youtube.com/watch?v=FIKNriGpncU&t=253s\

Q. Explain View in android.

Ans :-

- There are four basic views in android.
  - ➔ **Button**
  - ➔ **TextView.**
  - ➔ **ImageView.**
  - ➔ **EditText.**

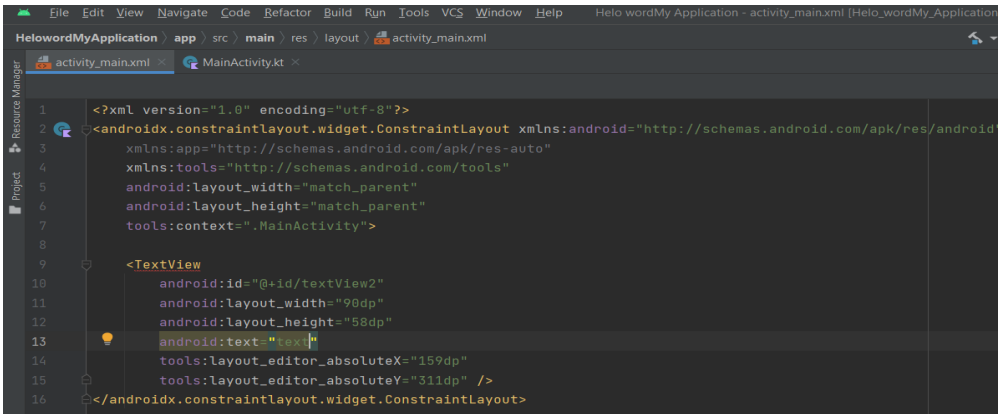


1) Text view :-

- TextView is a user interface control that is used to set and display the text to the user based on our requirements.
- The TextView control will act as like label control and it won't allow users to edit the text.
- A TextView is a complete text editor, however the basic class is configured to not allow editing.
- Label means textview in android.

Textview attributes

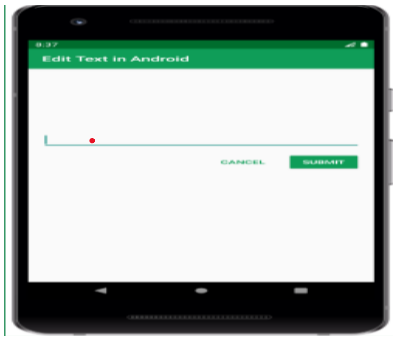
- **android: id** :- This is the ID which uniquely identifies the control **.**
- **android:capitalize:-** If set, specifies that this TextView has a textual input method and should automatically capitalize what the user types.
- **android:fontFamily**:- Font family (named by string) for the text.
- **android:maxHeight**:- deifne height of this view
- **android:maxwidth**:- deifne width of this view
- **android:text:-** text to be display is this text view.
- **Android text size:-** define the size of text to be display in text view.
- **Tools:layout_editor_absoluteX:-** define position in x axis .
- **Tools:layout_editor_absolutey:-** define position in y axis la



2) <u>Edit text</u>:-

- EditText is a predefined subclass of TextView.
- We use EditText in our applications in order to provide an input or text field specially in forms.

- EditText (or **PlainText**) is used in the app whenever you need input from the user side and proceed with its text (or value) in your app.
- EditText (or **<u>PlainText</u>**) is not only used to get plain text in your application, but even you can use it to get values such as email, number, password, etc.




```
<EditText
    android:id="@+id/username"
    android:layout_width="fill_parent"
    android:layout_height="wrap_content"
    android:hint="First Name"
    android:inputType="text"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent" />
```

Attributes of edittext:-

- **android:text**:- Used to set the text of the EditText
- **android:textSize:-** Used to set size of the text.
- **android:textColor:-** Used to set color of the text.
- **android:textStyle**:- Used to set style of the text. For example, bold, italic, bolditalic, etc.

3) Button view or Button:-

- In Android applications, a **Button** is a user interface that is used to perform some action when clicked or tapped.
- Android Button represents a push-button.
- Button is subclass of TextView class and CompoundButton is the subclass of Button class.
- There are different types of buttons in android such as RadioButton, ToggleButton, CompoundButton etc.

ATTRIBUTE AND DESCRIPTION

➔ **android:id:-**

- Used to specify the id of the view.

➔ **android:text:-**

- Used to the display text of the button.

➔ **android:textColor:-**

- Used to the display color of the text.

➔ **android:textSize:-**

- Used to the display size of the text.

➔ **android:background:-**

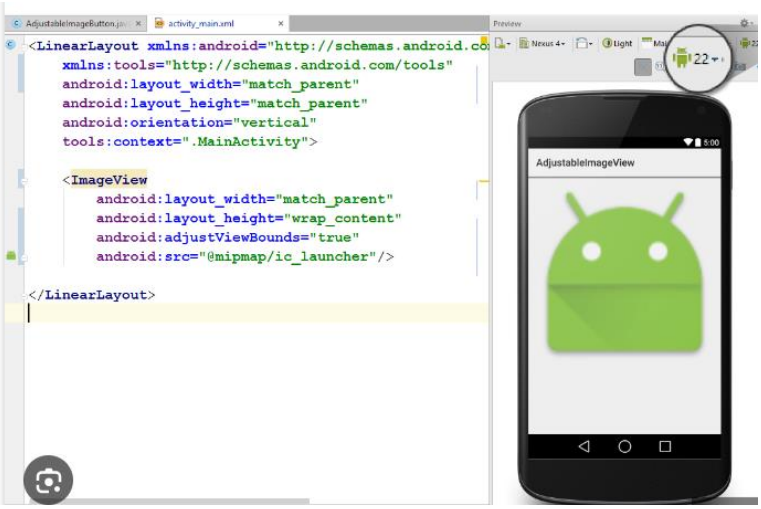
- Used to set the background of the view.

➔ **android:textStyle:-**

- Used to the display style of the text like Bold, Italic, etc.

4) IMAGE VIEW:-

- ImageView class is used to display any kind of image resource in the android application either it can be android. graphics. Bitmap or android. Graphics.
- ImageView is also used to control the size and movement of an image.



ATTRIBUTE AND DESCRIPTION

➔ **android:id:-**

- To uniquely identify an image view.

➔ **android:src/app:srcCompat:-**

- To add the file path of the inserted image.

➔ **android:background:-**

- To provide a background color to the inserted image.

➔ **android:layout_width:-**

- To set the width of the image.

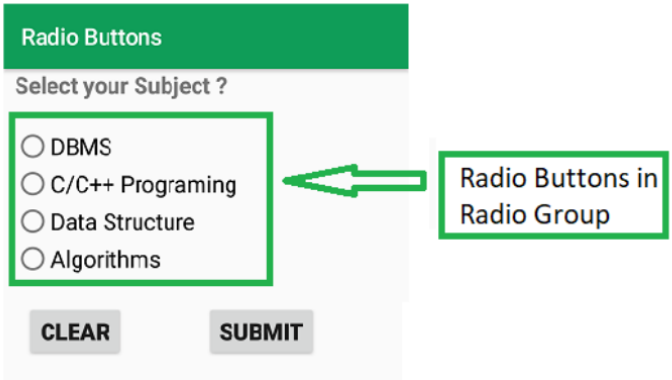
➔ **android:layout_height:-**

- To set the height of the image.

Q- EXPLAIN RADIO BUTTON.

- **RadioButton** is a two states button which is either checked or unchecked.
- If a single radio button is unchecked, we can click it to make checked radio button.

- Once a radio button is checked, it cannot be marked as unchecked by user.
- RadioButton is generally used with *RadioGroup*.
- RadioGroup contains several radio buttons, marking one radio button as checked makes all other radio buttons as unchecked.

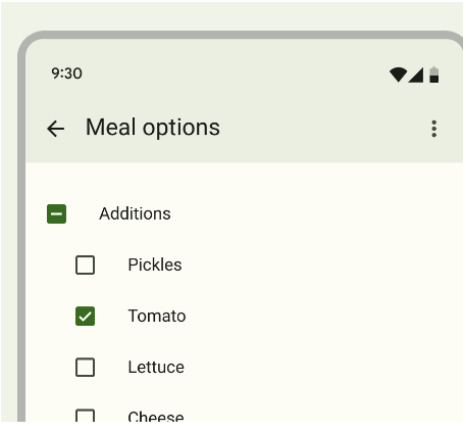


ATTRIBUTE AND DESCRIPTION

- ➔ **android:id:-** It is used to uniquely identify the control.
- ➔ **android:checked:-** It is used to specify the current state of radio button
- ➔ **android:gravity:-** It is used to specify how to align the text like left, right, center, top, etc.
- ➔ **android:text:-** It is used to set the text for the radio button.
- ➔ **android:textColor:-** It is used to change the color of text.
- ➔ **android:textSize:-** It is used to specify the size of the text.
- ➔ **android:textStyle:-** It is used to change the style (bold, italic, bolditalic) of text.
- ➔ **android:background:-** It is used to set the background color for radio button control.

Q- EXPLAIN CHECKBOX

- A CheckBox is a special kind of button in Android which has two states either checked or unchecked.
- There are many other uses of the CheckBox like offering a list of options to the user to choose from and the options are mutually exclusive i.e., the user can select more than one option.

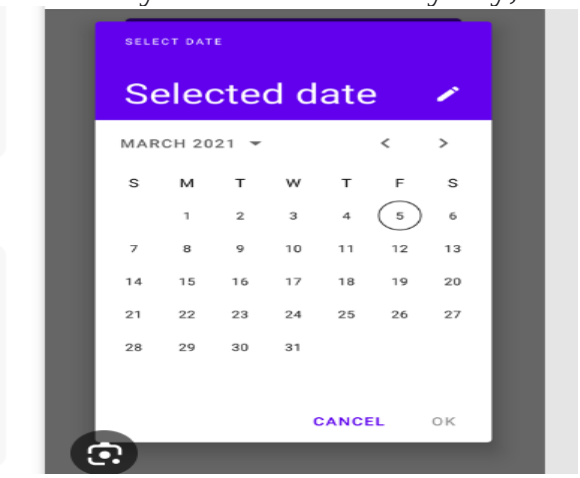


ATTRIBUTE AND DESCRIPTION

- **android:id:-** It is used to uniquely identify the control.
- **android:checked:-** To set the default state of a CheckBox as checked or unchechek
- **android:background**: To set the background color of a CheckBox
- **android:text**:- Used to store a text inside the CheckBox
- **android:fontFamily**:- To set the font of the text of the CheckBox
- **android:textSize**:- To set the CheckBox text size
- **android:layout_width**:- To set the CheckBox width.
- **android:layout_height**:- To set the CheckBox height

Q- EXPLAIN Date Picker.

- Android DatePicker is a widget to select date.
- It allows you to select date by day, month and year.



Method and description.

➔ **getDayOfMonth()**

- This method gets the selected day of month

➔ **getMonth()**

- This method gets the selected month

➔ **getYear()**

- This method gets the selected year

➔ **setMaxDate(long maxDate)**

- This method sets the maximal date supported by this DatePicker in milliseconds since January 1, 1970 00:00:00 in getDefault() time zone

➔ **setMinDate(long minDate)**

- This method sets the minimal date supported by this NumberPicker in milliseconds since January 1, 1970 00:00:00 in getDefault() time zone

➔ **getCalendarView()**

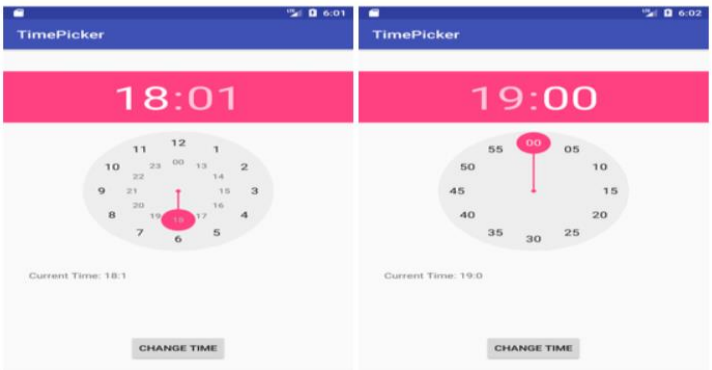
- This method returns calendar view

➔ **getFirstDayOfWeek()**

- This Method returns first day of the week

Q- EXPLAIN Time Picker.

- Time picker is a graphical user interface.
- It allows the user to select time from a time range.
- **Android TimePicker** widget is used to select date
- It allows you to select time by hour and minute. You cannot select time by seconds.
- TimePicker Dialog is used in many applications where we have to book appointments based on a specific time

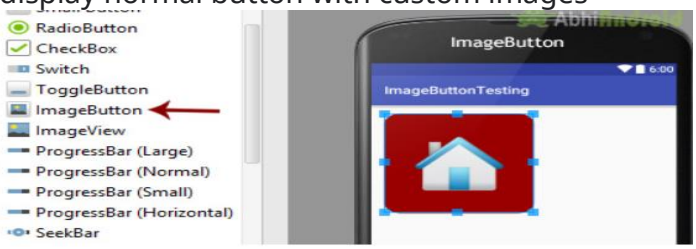


Method and description.

- **is24HourView()**
  ➔ This method returns true if this is in 24 hour view else false
- **isEnabled()**
  ➔ This method returns the enabled status for this view
- **setCurrentHour(Integer currentHour)**
  ➔ This method sets the current hour
- **setCurrentMinute(Integer currentMinute)**
  ➔ This method sets the current minute
- **setIs24HourView(Boolean is24HourView)**
  ➔ This method set whether in 24 hour or AM/PM mode

Q- EXPLAIN Image button.

- Image button is button that display image on button.
- Displays a button with an image (instead of text) that can be pressed or clicked by the user.
- By default, an ImageButton looks like a regular `Button`, with the standard button background that changes color during different button states.
- It display normal button with custom images



ATTRIBUTE AND DESCRIPTION

➔ **android:cropToPadding**

- If true, the image will be cropped to fit within its padding.

➔ **android:src**

- This sets a drawable as the content of this ImageView.

➔ **android:background**

- This is a drawable to use as the background.

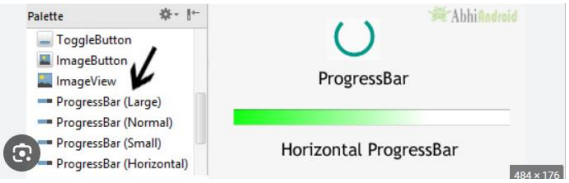
➔ **android:contentDescription**

- This defines text that briefly describes content of the view.

➔ **android:id**

- This supplies an identifier name for this view

Q- Explain progress bar.



- ProgressBars are used as loading indicators in android applications.
- These are generally used when the application is loading the data from the server or database.
- There are different types of progress bars used within the android application as loading indicators
- We can display the **android progress bar** dialog box to display the status of work being done e.g. downloading file, analyzing status of work etc.
- The ProgressDialog class provides methods to work on progress bar like setProgress(), setMessage(), setProgressStyle(), setMax(), show() etc. The progress range of Progress Dialog is 0 to 10000.

ATTRIBUTE AND DESCRIPTION

- **id:** id is an attribute used to uniquely identify a Progress bar.
- **max:** max is an attribute used in android to define maximum value of the progress can take. It must be an integer value like 100, 200 etc.
- **progress:** progress is an attribute used in android to define the default progress value between 0 and max. It must be an integer value.

- **background:** background attribute is used to set the background of a Progress bar. We can set a color or a drawable in the background of a Progress bar.

Q- EXPLAIN List view.

- ListView is widely used in android applications.
- A list is any information display or organized in logical form**.**
- It is view which group many item and display them in vertical scrollable list**.**
- A very common example of ListView is your phone contact book,
- List of scrollable items can be displayed in Android using ListView.
- it helps you to displaying the data in the form of a scrollable list.
- Users can then select any list item by clicking on it.
- ListView is default scrollable so we do not need to use scroll View or anything else with ListView.

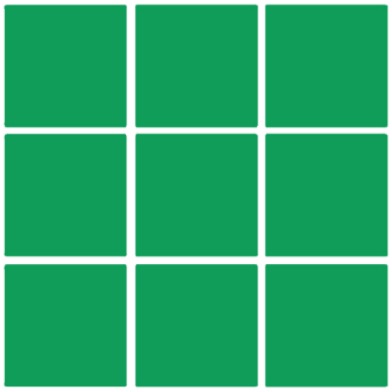
ATTRIBUTE AND DESCRIPTION

- **android:id**
  ➔ This is the ID which uniquely identifies the layout.
- **android:divider**
  ➔ This is drawable or color to draw between list items.

Q- EXPLAIN Grid view.

- In android GridView is a view group that display items in two dimensional scrolling grid (rows and columns),
- the grid items are not necessarily predetermined but they are automatically inserted to the layout using a ListAdapter.

- Users can then select any grid item by clicking on it. GridView is default scrollable so we don't need to use ScrollView or anything else with GridView.
- GridView is widely used in android applications. An example of GridView is your default Gallery, where you have number of images displayed using grid.
- Gridview is present inside Containers. From there you can drag and drop on virtual mobile screen to create it. Alternatively you can also XML code to create it.



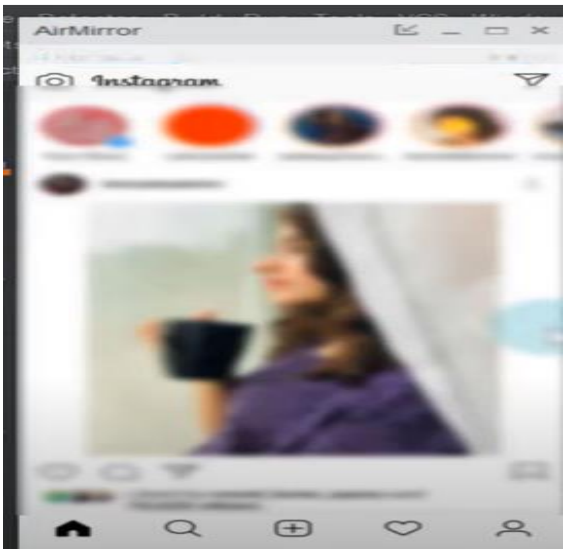
ATTRIBUTE AND DESCRIPTION

- **android:numColumns**: This attribute of GridView will be used to decide the number of columns that are to be displayed in Grid.
- **android:horizontalSpacing**: This attribute is used to define the spacing between two columns of GridView.
- **android:verticalSpacing**: This attribute is used to specify the spacing between two rows of GridView.
- **android:id**: This is the ID which uniquely identifies the layout.
- **android:columnWidth**: This specifies the fixed width for each column.

Q- EXPLAIN fragment.

- **Android Fragment** is the part of activity, it is also known as sub-activity.
- There can be more than one fragment in an activity. Fragments represent multiple screen inside one activity.

- Android fragment lifecycle is affected by activity lifecycle because fragments are included in activity.
- The **FragmentManager** class is responsible to make interaction between fragment objects.
- It is the modular section of the android activity that is very helpful in creating UI designs that are flexible in nature and auto-adjustable based on the device screen size.



Types of Android Fragments

- **Single Fragment:** Display only one single view on the device screen. This type of fragment is mostly used for mobile phones.
- **List Fragment:** This Fragment is used to display a list-view from which the user can select the desired sub-activity.

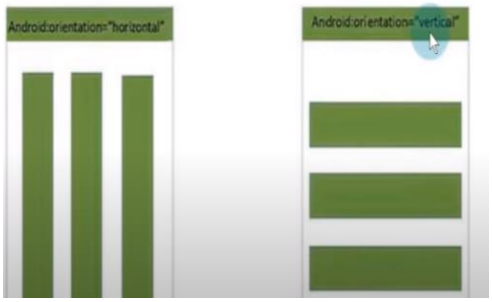
Q- what is layout ? How many types of Layouts are available in android?

Ans.

- A layout defines the structure for a user interface in your app, such as in an activity.
- Layout basically refers to the arrangement of elements on a page these elements are likely to be images, texts or styles.
- Relative Layout, Linear Layout, Table Layout etc

Q. EXPLAIN liner layout.

- We use this layout to place the elements in a linear manner
- A Linear manner means one element per line.
- LinearLayout is a view group that aligns all children in a single direction, vertically or horizontally.
- we can specify the layout direction with the android:orientation attribute.

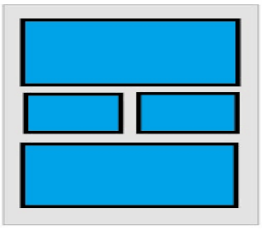


- there are two category of liner layout like vertically and horizontal.
  ➔ **Attributes**

- **Android:id-** this is id which is define uniquely the layout.
- **Android**:**divider:-** We use a color value in form of #RGB or many more.
- **Android**:**Gravity**:- this specifies how an object should position its content, on both X andY axes. Possible values are top bottom left right.

- **Android:orientation:-** this specifies direction of arrangement. By default is horizontal.

Q- Explain Relative layout.

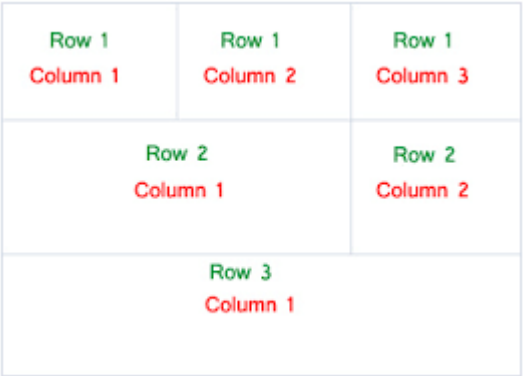


- A RelativeLayout is a very powerful utility for designing a user interface because it can eliminate nested view groups and keep your layout hierarchy flat, which improves performance.
- If you find yourself using several nested LinearLayout groups, you may be able to replace them with a single RelativeLayout .

➔ **Attributes:-**

- **android:id:** It uniquely identifies the Android Layout.
- **android:layout_height:** It sets the height of the layout.
- **android:layout_width:** It sets the width of the layout.
- **android:layout_gravity:** It sets the position of the child view.

Q- explain table layout.



- TableLayout is a ViewGroup subclass, used to display the child View elements in rows and columns.

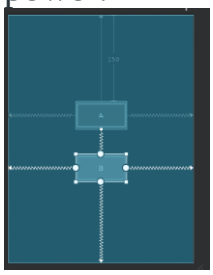
➔ **Attributes:-**

- **android:id:** It uniquely identifies the Android Layout.

- **android:layout_height:** It sets the height of the layout.
- **android:layout_width:** It sets the width of the layout.
- **android:layout_gravity:** It sets the position of the child view.

Q-explain constraint layout

- most of application use this types of layout.
- A ConstraintLayout is similar to a RelativeLayout, but having more power.



➔ **Attributes:-**

- **android:id:** It uniquely identifies the Android Layout.
- **android:layout_height:** It sets the height of the layout.
- **android:layout_width:** It sets the width of the layout.
- **android:layout_gravity:** It sets the position of the child view.

Q- Explain dialogs box or explain alert dialog box in android.

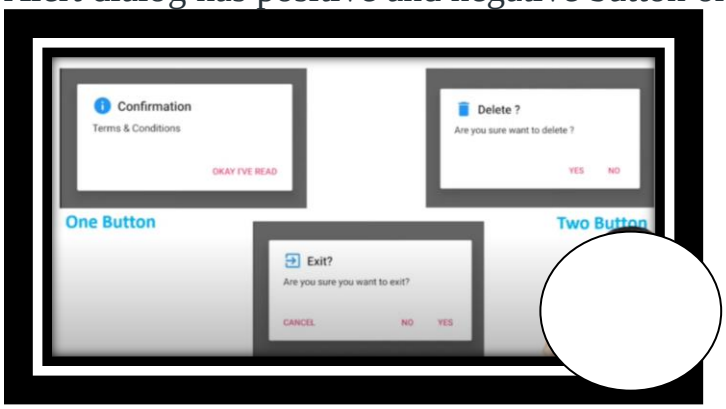
- A dialog is a small window that prompts the user to make a decision or enter additional information.
- A dialog doesn't fill the screen and is normally used for modal events that require users to take an action before they can proceed.

➔ **Alert dialog box:-**

- They are used to help user answer question, make selection, confirm actions and read warning or error message.
- Alert Dialog shows the Alert message and gives the answer in the form of yes or no.
- Alert Dialog displays the message to warn you and then according to your response, the next step is processed.
- Android Alert Dialog is built with the use of three fields: **Title, Message area, and Action Button.**

Alert Dialog code has three methods:

- **setTitle()** method for displaying the Alert Dialog box Title
- **setMessage()** method for displaying the message
- **setIcon()** method is used to set the icon on the Alert dialog box.
- There are three types of alert box like one button , two button, and three button.
- Alert dialog has positive and negative button or message view.



Q- Explain animation in detail.

- Animation is the process of creating motion and shape change.
- These are used when the ui change state, such as when new content loads.
- Animation also add a polished look to our app,which give it higher quality look and feel.
- We can change image move, move image, size increase and decrease of image.
  - ➔ **Drawing animation and working with animation**
- There are two types of drawing and working animation like **frame by frame animation** and **twined animation.**
  - ➔ **<u>Frame by frame animation:-</u>** in this we can swapping frame repeatedly so it appears continuous to the human eyes.
- Using this we can swap image and change the position of image.
- Its working in only image.

- W can not crate our shape and all other.

➔ **Twined animation:-**

- A tween animation can perform a series of simple transformations (position, size, rotation, and transparency) on the contents of a View object.
- So, if you have a TextView object, you can move, rotate, grow, or shrink the text. If it has a background image, the background image will be transformed along with the text**Types of animation**

➔ **Other types of animation**

- **Scale animation:-** used to make a smaller or larger view either on x and y axis.**(Small to big or big to small)**
- **Rotate animation:-** it is used to rotate view around pivot point a certain number of degree. **(rotate clock and anti clock wise)**
- **Translate animation:-**used to move view along the x or y axis. **(Marquee type).**
- **Alpha animation:-** transparency of view can be changed.**(visible 1 or un visible 0)**

➔ **Attribute**

- **Android:duration=1000(time in mili second)** :- the duration in which animation is completed.
- **Android:fillafter=true:-** define whether the view should be visible or not at the end of the animation.
- **Android:startoffset:-** refer the waiting time before an animation start.**(if we give 2 second time so animation display and start after 2 second .waiting time given)**
- **Android:** interpolator:- refer to rate of change in animation.**(speed of changing animation is speed or slow).**
- **Android:** repeatmode: mode of repetition.**(cycle or uncycle) .**

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Mobile Application Development in
Android using Kotlin
BCA & BSC IT SEM 6

UNIT – 4 Data Base Connectivity Using Sqlite

- Using Android Data and Storage APIs
- Managing data using SQLite.
- Sharing Data Between Applications with Content Providers.

Q-What is storage ?

- Storage is where you keep data, like music and photos.
- Memory is where you run programs, like apps and the Android system.

Q- Give the full form of API.

- Application Programming Interface.

Q-2. Explain using android data and storage api.

- Android provides many kinds of storage for applications to store their data.
- These storage places are
  ➔shared preferences,
  ➔internal Storage ,
  ➔External storage,
  ➔SQLite storage,
  ➔storage via network connection

➔ internal storage:-

- **Android Internal Storage** is the device memory in which we store the files.
- The file stored in the internal storage is **private in default**, and only the same application accesses it.
- They cannot be accessed from outside the application

- To read and write the data from (into) the file, Android provides **openFileInput()** and **openFileOutput()** methods.

➔ Write to File in Internal Storage:-

- To write the file in internal storage of device, **java.io** package offers **openFileOutput()** method.
- which returns the instance of **FileOutputStream** class.
- To write the data into the file call the **FileOutputStream .write()** method.

➔ Shared Preferences:-

- Android provides many ways of storing app data such as an SQLiteDatabase, saving a data text file and many more.
- One of the way to store data is called SharedPreference
- It store & retrieve data in the form of key and value pair.
- simple method to read and write them.
- Short data or singular data or one user data can perfect store in shared preference.

➔ **Method of shared preference**

- Apply():- it is abstract method.(**using .apply() method.**)
- Clear():- it will remove all the value from the editor.
- Remove():- it will remove particular values.
- Putstring(string key ,string value):- it will save float value.

- putINT(stringkey,int value):- it will save integer value.
- putLong:- it will save long values.
- Delete file():- delete a file saved on the internal storage.

➔ External Storage:-

- External storage such as a SD card, is a removable storage option for mobile phones that can be easily swapped out and replaced with a larger capacity card.
- It is typically slower than internal storage, which is built into the device and cannot be removed.

➔ Sqlite Storege:-

- Store structured data in private data base.
- SQLite is a opensource SQL database that stores data to a text file on a device.
- Android comes in with built in SQLite database implementation.
- SQLite supports all the relational database features.
- In order to access this database, you don't need to establish any kind of connections for it like JDBC,ODBC e.t.c.

➔ Storage Via network connection:-

- Store data on the web with our own network server.
- Network-attached storage (NAS) devices store and share data for multiple computers, that can be accessed remotely.

Q- explain SQLITE in detail.

- SQLite is another data storage available in Android where we can store data in the user's device and can use it any time when required.
- SQLite is an open-source relational database.
- It used to perform database operations on android devices such as storing, manipulating or retrieving data from the database.
- It is embedded in android by default. So, there is no need to perform any database setup or administration task.

- This is a series of 4 articles in which we are going to perform the basic **CRUD (Create, Read, Update, and Delete)** operation with **SQLite Database** in Android.
- We can perform so many operations on this data such as adding new data, updating, reading, and deleting this data.
- Data is stored in the SQLite database in the form of **tables.**
- When we stored this data in our SQLite database it is arranged in the form of tables that are similar to that of an excel sheet.
- The **SQLiteOpenHelper** class provides the functionality to use the **SQLite** database.

➔ SQLiteOpenHelper class:-

- The **android.database.sqlite.SQLiteOpenHelper** class is used for database creation and version management.
- For performing any database operation, you have to provide the implementation of **onCreate()** and **onUpgrade()** methods of SQLiteOpenHelper class.

➔ Important Methods in SQLite Database

- **getColumnNames():-** This method is used to get the Array of column names of our SQLite table.
- **getCount():-**This method will return the number of rows in the cursor.
- **isClosed():-**This method returns a Boolean value when our cursor is closed.
- **getColumnCount():-** This method returns the total number of columns present in our table.
- **getPosition():-**This method will return the current position of our cursor in our table.

➔ Query in sqlite:-

- **Create :-** SQLite **CREATE TABLE** statement is used to create a new table in any of the given database.
- CREATE TABLE database_name.table_name(
  column1 datatype PRIMARY KEY(one or more columns),

```
column2 datatype,
column3 datatype,
.....
columnN datatype
);
```

```
sqlite> CREATE TABLE COMPANY(
   ID INT PRIMARY KEY     NOT NULL,
   NAME          TEXT    NOT NULL,
   AGE           INT     NOT NULL,
   ADDRESS       CHAR(50),
   SALARY        REAL
);
```

- **Insert :-** SQLite **INSERT INTO** Statement is used to add new rows of data into a table in the database.

Syntax

- INSERT INTO TABLE_NAME [(column1, column2, column3,...columnN)] VALUES (value1, value2, value3,...valueN);

Example

```
sqlite> CREATE TABLE COMPANY(
   ID INT PRIMARY KEY     ,
   NAME          TEXT    NOT NULL,
   AGE           INT     NOT NULL,
   ADDRESS       CHAR(50),
   SALARY        REAL
);
```

```
INSERT INTO COMPANY (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (1, 'Paul', 32, 'California', 20000.00 );

INSERT INTO COMPANY (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (2, 'Allen', 25, 'Texas', 15000.00 );

INSERT INTO COMPANY (ID,NAME,AGE,ADDRESS,SALARY)
VALUES (3, 'Teddy', 23, 'Norway', 20000.00 );
```

- **Select :-** SQLite **SELECT** statement is used to fetch the data from a SQLite database table.
- **Syntax:**-
- SELECT column1, column2, columnN FROM table_name;
  Ex. SELECT * FROM table_name;
- **Update:**- SQLite **UPDATE** Query is used to modify the existing records in a table. You can use WHERE clause with UPDATE query to update selected rows, otherwise all the rows would be updated.

Syntax:-

➔ UPDATE table_name
SET column1 = value1, column2 = value2...., columnN = valueN
WHERE [condition];

Example:-

```
UPDATE COMPANY SET ADDRESS = 'Texas' WHERE ID = 6;
```

Delete :- SQLite **DELETE** Query is used to delete the existing records from a table.

Syntax:- DELETE FROM table_name
WHERE [condition];.

```
DELETE FROM COMPANY WHERE ID = 7;
```

Q. Explain content provider.

- In Android, Content Providers are a very important component that serves the purpose of a relational database to store the data of applications.
- The role of the content provider in the android system is like a central storage(container) in which data of the applications are stored, and it facilitates other applications to securely access and modifies that data based on the user requirements.
- Android system allows the content provider to store the application data in several ways.
- Users can manage to store the application data like images, audio, videos, and personal contact information by storing them in **SQLite Database, in files**, **or even on a network**.

- In order to share the data, content providers have certain permissions that are used to grant or restrict the rights to other applications to interfere with the data.

➔ **Content URI**

- **Content URI(Uniform Resource Identifier)** is the key concept of Content providers.
- To access the data from a content provider, URI is used as a query string.
- Content URI has many part likes:- content, authority , optionalpath,optionalid.
- There are four types of operation in content provider like **CRUD Create, read, update and delete.**
- it provide many method like query() , insert(), update(),delete(),gettype(), Oncreate().

Q. Explain Curosr.

→ cursor is used for query result.

➔ Cursor is position indicator.

➔ It is object that represent a bunch of rows in database.

SELECT * FROM pets;

_id	name	breed	gender
1	Toto	Terrier	0
2	Binx	Bombay	0
3	Lady	Cocker Spaniel	1
4	Cat	Tabby	0
5	Baxter	Border Terrier	0
6	Arlene	Unknown	1

.

➔

SELECT * FROM pets WHERE name = "Toto";

_id	name	breed	gender	weight
1	Toto	Terrier	0	14

** cursor metod:-

1) Movetofirst():- move the cursor to the first row.
2) Movetolast():- move the cursor to the last row.
3) Movetonext():- move the cursor to next row.
4) Movetoposition():- move the cursor to absolute position.
5) Movetoprevious():- move the cursor to previous row.

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Mobile Application Development in Android using Kotlin
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UNIT – 5 Location based services and common API

- **Using Global Positioning Services (GPS)**
- **Geocoding Locations**
- **Mapping Locations**
- **Many more with location based services**
- **Android networking API**
- **Android web API**
- **Android telephony API**
- **Notifying the user, Notifying with the status bar**
- **Vibrating the phone**
- **Blinking the lights**
- **Customizing the notifications Services**
- **Application development using JSON in MySQL**
- **Publish android application.**

Q-1.Using Global positioning services or explain GPS in detail.

- Global Positioning Systems, widely known as GPS.
- Gps work through the space.
- It capture **3d data for** various application.
- Basically gps is found for military department.
- A GPS system consists of a network of 24 orbiting satellites,It is called NAVSTAR.(Navigation System with Time and Ranging) and placed in space in six different orbital paths with four satellites in each orbital.
- The orbit period of these satellites are 12 hours.
- The satellite signals can be received anywhere and at any time in the world.
- The first GPS satellite was launched in February 1978.
- The speed of this satellite is 7000 mile per hours.
- It continues move in earth .
- Each satellite weight are approx. 862 kg.
- Maintenance are continuously required for all these satellite.
- Approx. 7 or 8 year life of all these satellite.
- .

➔ **History of the Global Positioning System (GPS)**

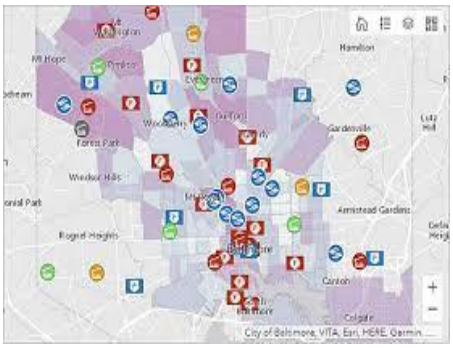
- 1957 - Soviet Union launched Sputnik I satellite.
- 1960s - U.S. Navy tracked U.S. submarines with satellite navigation.
- 1978 - NAVSTAR Block I GPS satellite was launched.
- 1983 - U.S. announced that it will make GPS available for civilian use after Korean Air flight 007 shut down.
- 1989 - Magellan introduced NAV 1000, the first hand-held GPS device. First Block II satellite is launched.
- 1993 - Constellation of 24 satellite system became operational.
- 1995 - Full Operational Capability (FOC) declared.
- 2000 - Selective Availability discontinued by the U.S. Government.

- 2004 - Qualcomm successfully completed test of live assisted GPS on a mobile phone.
- 2008 - Block II satellite launched.
- 2016 - GPS IIF satellite launched.

➔ **Benefits of GPS**

- At the start, more than 1000 portable commercial GPS receivers were purchased for military use.
- They were carried by soldiers and attached to vehicles, helicopters, and aircraft instrument panels.
- GPS receivers were used in several aircrafts, including B-2 bombers and F-16 fighters etc.
- Now a days gps use in many sector like In gas, in forest , in oli and petrol , in sports , also use farmer , sports mans and many more.

Q-2. Explain geocoding location in detail.



- The full form of Geocoding is **geographical coordinates.**
- Geocoding is the process of translating a physical address into a geographical location.
- Geocoding is subset of GIS(geographical information system).
- Address geocoding, or simply geocoding, is the process of taking a text-based description of a location, such as an address or the name of a place, and returning geographic coordinates, frequently latitude(અક્ષાંસ)/longitude(રેખાંશ) pair, to identify a location on the Earth's surface.

- Geocode is a code that represents a geographic entity (location or object).
- **Geocoder** is a piece of software or a (web) service that implements a geocoding process.
- The first geocoding practices were implemented in 1960 with the invention of the Canada Geographic Information System (CGIS), which mapped, stored, and analyzed information about wildlife, forestry, and agriculture.
- The geocoding is two types of process first is proper address can convert into latitude and longitude. And other is convert latitude and longitude into a human readable form.

- Using geocoding we can show a famous place near our find out location.(like temple, shop, hotel etc.)

- The end of the 20th century had seen geocoding become more user-oriented, specially via open-source GIS software.

Q-3 Explain mapping location. Or explain google map.

- Android provides facility to integrate Google map in our application.
- Google map displays your current location, navigate location direction, search location etc.
- We can also customize Google map according to our requirement.

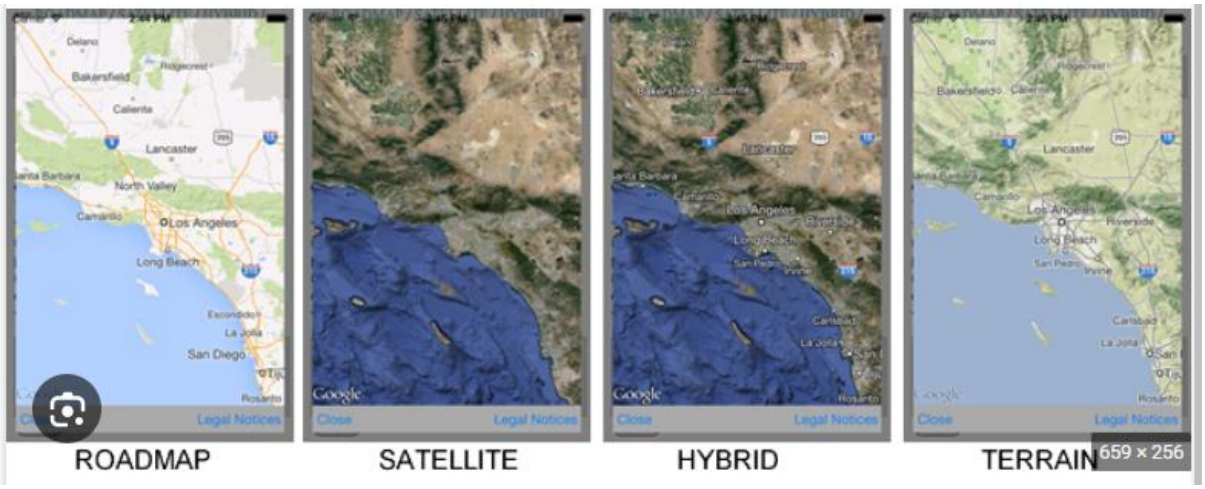
➔ **<u>Types of Google Maps:-</u>**

- There are four different types of Google maps.
- These maps are as follow:
- **A) Normal:** This type of map displays typical road map, natural features like river and some features build by humans.

b) Hybrid: This type of map displays satellite photograph data with typical road maps. It also displays road and feature labels.

c) Satellite: Satellite type displays satellite photograph data, but doesn't display road and feature labels.

d) Terrain: This type displays photographic data. This includes colors, lines and labels and perspective shading.



- **Method of google map.**
  1) addCircle:- This method add circle to map.
  2) addPolygon:-This method add polygon to map.
  3) clear():- This method removes everything from the map.

Q-4 Explain Network API.(api= Application programming interface)

- Android network connectivity services allow us to check the network connectivity information of the device.
- Android lets your application connect to the internet or any other local network and allows you to perform network operations.
- A device can have various types of network connections likes wifi, mobile net and blue tooth.
- It is very important to check the internet connection of the device while performing the task which is based on internet such as fetching data from the server (internet) or writing data to the server.
- To access the network connectivity of a device, we need to provide the network access permission in AndroidMenifest.xml file.
- In network api we perform the following steps likes checking network connection , performing network operation.

i) Checking network connection:-

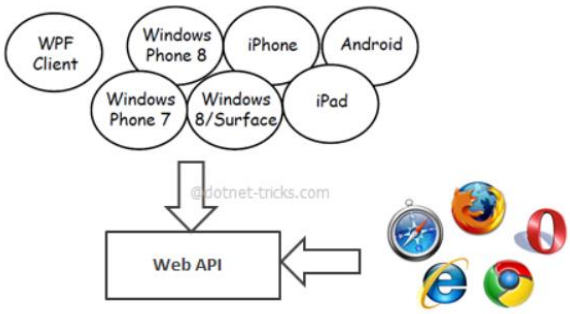
- Before you perform any network operations, you must first check that are you connected to that network or internet.
- For this android provides ConnectivityManager class.
- You need to create an object of this class by calling getSystemService() method.

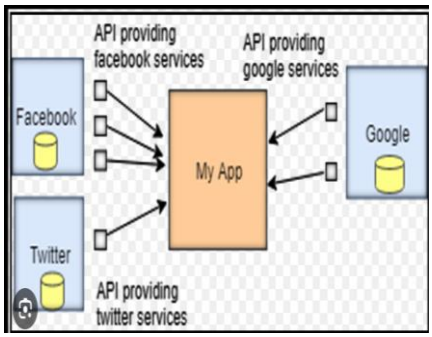
ii) Performing network operation:-

- After checking that you are connected to the internet, you can perform any network operation.
- we are also fetching the website from a url.
- Android provides HttpURLConnection and URL class to handle these operations.

Q-5 Explain web Api.(application programming interface)

- A Web API is an application programming interface for the Web.
- Web api is bridge between browser and device.
- web api used in two ways like browser api and server api.
- A Browser API can extend the functionality of a web browser.
- All browsers have a set of built-in Web APIs to support complex operations, and to help accessing data.
- A Server API can extend the functionality of a web server.
- web api is used to display web page in android.
- The web page can be loaded from same application or URL.
- Android Web api use webkit engine to display web page.
- A web API works as a middleman between the applications and business servers.





Q-6.explain telephony API.

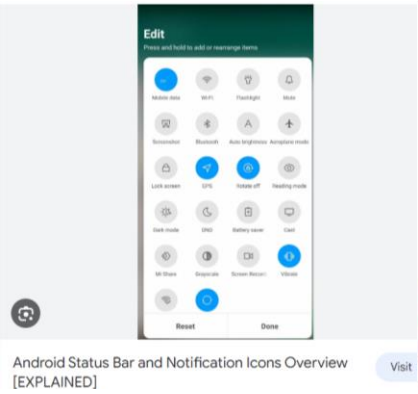
- The telephony system/API is a software framework to provide mobile phones with telephony functionalities, such as voice call, Video call, SMS,MMS ,data service, network management and so on.
- Telephony describes the technology that allows people to interact and communicate across long distances through the electronic transmission of voice, fax or other information.
- The Telephony Application Programming Interface (TAPI)
- TAPI was introduced in 1993 as the result of joint development by Microsoft and Intel.

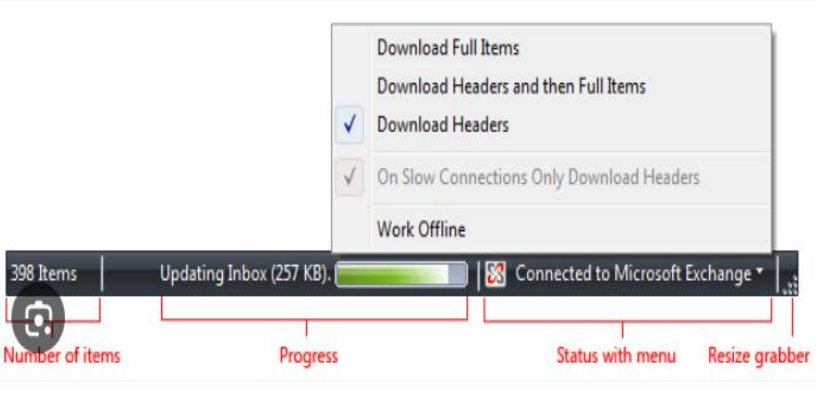
➔ What are the uses of telephony?

- Making international/national calls.
- Sending Short Message Service (SMS) text messages.
- Sending or receiving banking documents via fax.
- Conducting video and audio calls.
- Using communication software that can integrate with other business tools.
- Group or team messaging.
- File sharing.

Q-7:- Explain Notifying the user, Notifying with the status bar

- A notification is a message, email, icon, or another symbol that appears when an application wants you to pay attention
- Notifications are a way to let you know that something new has happened so you don't miss anything that might be worth your attention and appears whether you are using an application or not.
- A status bar notification adds an icon to the system's status bar
- When the user selects the expanded message, Android fires an Intent that is defined by the notification (usually to launch an Activity).





Q-8 Vibrating the phone

- using vibrate service, we can vibrate android mobile.
- Making Android phones vibrate is a good way to provide feedback to users or to interact with users even when phone volume is low.
- Entry on Menifest file:-

  <uses-permission android:name="android.permission.VIBRATE>

Q-9) why JSON is used in android ?

- JSON stands for **J**ava**S**cript **O**bject **N**otation
- JSON is a lightweight format for storing and transporting data
- JSON is often used when data is sent from a server to a web page
- JSON is "self-describing" and easy to understand.
- JSON is a language-independent data format.
- It was derived from JavaScript, but many modern programming languages include code to generate and parse JSON-format data.

- JSON filenames use the extension .json.
- It is an alternative to XML.
- JSON is faster and easier than xml for AJAX applications.
- Unlike XML, it is shorter and quicker to read and write.
- Android supports all the JSON classes such as JSON Stringer, JSON Object, JSON Array, and all other forms to parse the JSON data and fetch the required information by the program.

Q-10. How to Publish android application. Or how to upload app in play store.

- Here give the few step for how to upload app on google play store?

Step 1: Create a Google Developer account:-
Step 2: Add a Merchant account:-
Step 3: Prepare the documents:-
Step 4: Study Google developer policies:-
Step 5: Technical requirements :-
Step 6: Creating the App on the Google Console:-
Step 7: Store listing:-
Step 8: Content Rating:-
Step 9: Pricing the Application:-
Step 10: Upload APK and Send it for Review:-

Step 1: Create a Google Developer account:-

- ➔ Creating a Google Developer account is something you can do at the beginning of the app development process.
- ➔ Without registering a Google Developer Account, you can't publish your app on the Play Market.
- ➔ You can use any of your current Google accounts or create another one to sign up for a Google Developer Account.
- ➔ It doesn't matter whether it's a private or corporate account.
- ➔ You may easily transfer your app to another one in the future.
- ➔ The creation process includes signing the Google Play Developer distribution agreement, adding some personal information, and paying a one-time registration fee of $25.

Step 2: Add a Merchant account:-

- If you plan to sell paid apps or in-app purchases, you have to create a Google Merchant Account.

- There you can manage app sales and your monthly payouts, as well as analyze sales reports.
- Once you finish creating the Merchant profile, the developer account gets automatically linked to it.

Step 3: Prepare the documents:-

- Paperwork always requires much effort, especially when it comes to any kind of legal documents.
- Based on our experience, we highly recommend starting to prepare the End User License Agreement (EULA) and Privacy Policy in advance.
- EULA is an agreement between you as an owner and a user of your product. In brief, it contains:

- What the users can do with the app, and what they aren't allowed to do
- Licensing fees.
- Make sure to include in the Privacy Policy

Step 4: Study Google developer policies:-

- We guess you already made up your product concept.
- Now it's time to make sure that every feature you will implement in the app is aligned with the Google Developer Policies.
- These documents explain how mobile apps need to be developed, updated, and promoted to support the store's high-quality standards.

Step 5: Technical requirements :-

- The package name should be suitable over the life of your application. You cannot change it after the distribution.
- Signed App Release With a Signing Certificate.
- Google set the limit size of the uploaded file: 100MB for Android 2.3 and higher (API level 9-10, 14 and higher) and 50MB for lower Android versions.

Step 6: Creating the App on the Google Console:-

- Now you have the file that is ready for uploading. It's time to get to the fun part. Let's create a new app in your Developer Account:

➔ Reach to All applications tab in the menu
➔ Now select Create Application
➔ Choose the app's default language from the drop-down menu
➔ Add a brief app description (you can change it later)

➔ Tap on Create

Step 7: Store listing:-

- It contains the most important information useful for app store optimization (ASO) and gives the users more details about your app before downloading.
- The mandatory sections are marked with *.
- You may need some designer and copywriter efforts, so it's better to start preparing the following materials in advance.

➔ Product description
➔ Screenshots
➔ Icon
➔ Feature graphic
➔ Promo video
➔ Tags
➔ Contact details
➔ Privacy Policy

Step 8: Content Rating:-

- In order not to be marked as an Unrated App (that may lead to app removal), pass a rating questionnaire.
- You can easily find this section on the left-side menu.
- The information provided in the questionnaire must be accurate.
- Any misrepresentation of your app's content might lead to suspension or removal of the Google Play account.

Step 9: Pricing the Application.

Step 10: Upload APK and Send it for Review.